The Ledger of Civilization:

A Theory of Reciprocal Relations

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Abstract

This paper develops the foundational framework on which a series of companion papers rests. Its central claim is that civilization is best understood not as a collection of monuments, institutions, or ideas, but as an institutionalized order of reciprocal relations among actors. These relations can be tracked on a *ledger* that records, for each relation, what each party contributes, what each party receives, what it costs each party to leave, and how much voice each party has. The paper proceeds from first principles. It defines the state of the ledger, derives the classification of relations into symbiotic, parasitic, and predatory from the sign and sustainability of net transfers, and formalizes the viability set, the injustice gap, the forces of drift, and the justice mechanism. It then derives, with proofs, the five propositions on which the companion papers depend: the extraction ceiling (bounded by exit costs), the patience threshold (the optimal host stock $K^*(r) = (1 - r)/2$), the legitimacy constraint (the cost of governance is decreasing in legitimacy), the coalitional instrument (the distribution is set by the weighting vector λ_i), and the dynamical system (drift versus justice). Each proposition is stated with its assumptions, proved, and followed by the corollaries the companion papers cite. The paper's central question, which the entire series inherits, is: *whose ledger is this, and for whom?*

Keywords: ledger, civilization, reciprocity, symbiotic, parasitic, predatory, exit costs, voice, legitimacy, injustice gap, justice mechanism, discount rate, extraction ceiling, weighting vector

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1 Introduction

Every civilization is, at its core, a set of relations. Who gives, who gains, who loses, and how these distributions are justified form the substance of social order [4, 2]. This set of relations is what this paper calls *the ledger*. The term is not merely metaphorical; it is structural. Every stable order institutionalizes some distribution of shares, obligations, and access to voice, while simultaneously imposing costs on the act of departure from its relations.

The standard story presents civilization as beginning with an agreement—a social contract, a covenant, a constitution—and as progressing through the accumulation of monuments, laws, and ideas. This paper argues that the standard story inverts the causal order. The monuments, laws, and ideas are *outcomes* of the ledger, not the ledger itself. The Pyramids prove nothing about justice; they prove that someone could command labor. To understand a civilization, one must look at the relations that produced its monuments, not at the monuments.

The purpose of this paper is to make the framework precise enough to be used. The companion papers—on politics, governance, government, democracy, religion, voice, race, and the state—all assume the definitions and propositions derived here. They cite them by number. This paper therefore does not merely state the framework; it derives it, so that the foundation is firm and the claims that rest on it are inspectable.

The paper proceeds as follows. Section 2 defines the formal setup: actors, relations, and the state of the ledger. Section 2.3 defines net gain, capacity, and the thresholds. Section 3 derives the classification of relations. Section 4 defines the viability set and the injustice gap. Section 5 formalizes drift and the justice mechanism, and derives the dynamical system. Section 6 derives the extraction ceiling. Section 7 derives the patience threshold. Section 8 derives the legitimacy constraint. Section 9 derives the coalitional instrument. Section 10 collects the core propositions. Section 11 concludes.

2 The Formal Setup

2.1 Actors and Relations

Definition 2.1 (Civilization)**.** *A civilization* Γ *is a pair* $\Gamma = (N, E)$*, where* N *is a set of actors (individuals, groups, classes, or institutions) and* E *is a set of institutionalized relations between them. A* relation *is any recurring, socially recognized connection that carries expectations of give and take [3, 4].*

In plain language: civilization is the web of relationships that hold people together through shared rules, norms, and obligations. A family, a firm, a state, and an empire are all ledgers of relations at different scales. Someone contributes labor, money, time, or loyalty. Someone receives the benefit. Someone can leave. Someone can speak.

For each relation $e \in E$, let $N(e) \subseteq N$ denote the set of actors party to e. The framework is agnostic about the cardinality of $N(e)$; the two-party case is the canonical one, but the definitions extend to relations among many parties.

2.2 The State of the Ledger

Definition 2.2 (State of the Ledger)**.** *The state of the ledger* S *is the collection*

$$S = \{(s_i^e,\ c_i^e,\ k_i^e,\ v_i^e) : i \in N,\ e \in E,\ i \in N(e)\}, \tag{1}$$

where, for each actor i in relation e:

- $s_i^e \in (0,1)$ *is the* share *of the relation's benefits accruing to actor i;*
- $c_i^e \geq 0$ *is the* contribution *of actor i to relation e (labor, care, resources, obedience, or loyalty);*
- $k_i^e \in (0,\infty)$ *is the* exit cost *for actor i to leave relation e (economic precarity, social ostracism, loss of community, legal penalties, or violence);*
- $v_i^e \in (0,1)$ *is the* voice *of actor i within relation e (their capacity to meaningfully influence the terms of that relation).*

The shares are normalized so that, for each relation e, the shares of the parties sum to at most one: $\sum_{i \in N(e)} s_i^e \leq 1$. The residual, if any, is the relation's unallocated surplus.

In plain language: for every relationship, the ledger records four things—what you get, what you give, what it costs to walk away, and how much say you have. These four numbers determine whether a relationship feels fair, tolerable, or intolerable.

2.3 Benefits and Net Gain

Definition 2.3 (Net Gain)**.** *Let $B^e \geq 0$ be the total benefit produced by relation e. The* benefit *accruing to actor i from relation e is $s_i^e B^e$. The* net gain *of actor i from relation e is*

$$n_i^e = s_i^e B^e - c_i^e. \tag{2}$$

The total net gain *of actor i is the sum over all relations in which i participates:*

$$n_i = \sum_{e : i \in N(e)} n_i^e. \tag{3}$$

Net gain is the primitive of the framework. It is the difference between what an actor receives and what an actor gives. A positive net gain means the actor is better off inside the relation than outside it; a negative net gain means the actor is being depleted.

In plain language: net gain is the bottom line of any relationship—what you get minus what you give. If it is positive, the relationship is working for you. If it is negative, you are being drained.

2.4 Capacity and Thresholds

Definition 2.4 (Capacity)**.** *The* capacity *of actor i, denoted $K_i \geq 0$, is the stock of resources—material, social, and psychological—that enables i to continue participating in the ledger. Capacity is reproduced by positive net gain and depleted by negative net gain:*

$$\frac{dK_i}{dt} = n_i. \tag{4}$$

Capacity is the bridge between the static ledger and its dynamics. An actor whose net gain is positive reproduces their capacity; an actor whose net gain is negative is slowly depleted. The classification of relations in Section 3 turns on whether capacity is reproduced, depleted-but-surviving, or destroyed.

In plain language: capacity is your ability to keep going. A relationship that gives you more than it takes keeps you going; a relationship that takes more than it gives slowly wears you down.

Definition 2.5 (Acceptance and Rupture Thresholds)**.** *For each actor i, there are two thresholds:*

- *the* acceptance threshold $\bar{a}_i$*: the minimum net gain n_i that actor i will tolerate without grievance;*
- *the* rupture threshold ρ_i*: the net gain below which actor i will rupture—exit the relation, or revolt against it.*

The thresholds are ordered: $\rho_i < \bar{a}_i$. An actor tolerates net gain between ρ_i and $\bar{a}_i$ (with grievance), and ruptures below ρ_i.

The ordering $\rho_i < \bar{a}_i$ encodes a familiar fact: people tolerate unfairness for a long time, but only up to a point. The gap between the two thresholds is the actor's tolerance for injustice.

In plain language: think of these as a person's breaking point. People will often tolerate unfairness for a long time, but only up to a point. The acceptance threshold is where the complaining starts; the rupture threshold is where the leaving starts.

3 The Classification of Relations

The classification of relations follows from the sign and sustainability of net transfers. Consider a two-party relation e between actors i and j.

Definition 3.1 (Symbiotic, Parasitic, and Predatory Relations)**.** *A relation e between i and j is:*

- symbiotic *if $n_i^e > 0$ and $n_j^e > 0$: both parties reproduce their capacity;*
- parasitic *if $n_i^e > 0$ and $n_j^e < 0$, but j's capacity K_j remains above the survival threshold $K_{\min}$: one party extracts from a surviving host;*
- predatory *if $n_i^e > 0$ and $n_j^e < 0$, and j's capacity K_j falls below the survival threshold $K_{\min}$: one party extracts by destroying the host's capacity to survive.*

The distinction between parasitic and predatory is not the sign of the transfer but its sustainability. A parasitic relation depletes the host but leaves it alive; a predatory relation depletes the host to the point of destruction. The same relation can be parasitic in the short run and predatory in the long run, if the depletion is sustained.

In plain language: a relationship is symbiotic when both people are better off in it; parasitic when one person is being drained but can still survive; predatory when one person is being drained to the point of destruction. The difference between parasitic and predatory is whether the host survives.

Proposition 3.1 (Exhaustiveness of the Classification)**.** *For any two-party relation e, exactly one of the following holds: e is symbiotic, e is parasitic, e is predatory, or e is degenerate (both parties have non-positive net gain).*

Proof. For each party, net gain is either positive, zero, or negative. The four cases are: (i) both positive (symbiotic); (ii) one positive, one negative, host survives (parasitic); (iii) one positive, one negative, host destroyed (predatory); (iv) neither positive (degenerate). The degenerate case—a relation that benefits no one—is unstable and dissolves, so it is excluded from the durable classification. The three durable types are exhaustive. □

Corollary 3.1 (Every Civilization Is a Mix)**.** *Every civilization is a mix of symbiotic, parasitic, and predatory relations. The question is never whether a civilization is good or bad in the abstract; it is which relations are symbiotic, which are parasitic, which are predatory, and who occupies each position.*

Proof. A civilization is a set of relations, and each relation falls into one of the three durable types. A single society may have symbiotic relations among its citizens, parasitic relations with its colonies, and predatory relations with its slaves. The mix is the civilization. □

4 The Viability Set and the Injustice Gap

4.1 The Injustice Gap

Definition 4.1 (Injustice Gap)**.** *The* injustice *experienced by actor i is the amount by which their net gain falls below their acceptance threshold:*

$$g_i = \max\big(0,\ \bar{a}_i - n_i\big). \tag{5}$$

The injustice gap *of the ledger is the aggregate:*

$$G(S) = \sum_{i \in N} g_i. \tag{6}$$

The injustice gap is zero when every actor receives at least their acceptance threshold, and positive when some actors are squeezed below it. It is the measure of how far the realized ledger is from the ledger the population would accept.

In plain language: the injustice gap is the distance between what people are actually getting and what they would accept. When it is zero, everyone is at least satisfied. When it is large, a lot of people are being squeezed.

4.2 The Viability Set

Definition 4.2 (Viability Set)**.** *The* viability set V *is the set of states S in which no actor is below their rupture threshold:*

$$V = \{S : n_i(S) \geq \rho_i \text{ for all } i \in N\}. \tag{7}$$

A state in V is one in which no actor has yet ruptured. A state outside V is one in which at least one actor has been pushed past their breaking point, and the order is in the process of rupturing.

In plain language: an order is viable when no one has been pushed past their breaking point. Once someone is pushed past it, the order starts to come apart.

Proposition 4.1 (The Injustice Threshold)**.** *There exists an* injustice threshold $\bar{G} > 0$ *such that the order is viable if and only if* $G(S) \leq \bar{G}$*. The threshold is the minimum injustice gap at which some actor reaches their rupture threshold:*

$$\bar{G} = \min_{S \in \partial V} G(S), \tag{8}$$

where ∂V *is the boundary of the viability set.*

Proof. The viability set V is defined by the constraints $n_i \geq \rho_i$. As the injustice gap $G(S)$ increases, actors are pushed toward their rupture thresholds. The first actor to reach their rupture threshold defines the boundary of V. The injustice gap at that boundary is $\bar{G}$. For $G(S) \leq \bar{G}$, no actor has ruptured, so the order is viable. For $G(S) > \bar{G}$, at least one actor has ruptured, so the order is not viable. □

Corollary 4.1 (Viability Is a Property of the Gap)**.** *The viability of an order is a property of the injustice gap, not of the order's formal structure. An order with a large injustice gap is unviable regardless of its constitutional form; an order with a small gap is viable regardless of its form.*

5 Drift and the Justice Mechanism

5.1 Drift

Definition 5.1 (Drift)**.** Drift *is the natural tendency of the ledger to become more unequal over time. Formally, drift is a positive rate* $d(G) > 0$ *at which the injustice gap grows in the absence of any corrective mechanism:*

$$\left.\frac{dG}{dt}\right|_{no\ justice} = d(G) > 0. \tag{9}$$

Drift has a structural source. The parties who benefit from the current allocation have both the incentive and the means to entrench their advantage: they accumulate resources, which buy voice, which lowers their exit costs, which raises the extraction ceiling on others. The result is a self-reinforcing tendency toward concentration. Drift is not a moral failure; it is the default direction of an unregulated ledger.

In plain language: systems left alone tend to become more unequal. The people who benefit from the current arrangement have both the motive and the means to entrench their advantage. Drift is the default direction of an unregulated ledger.

5.2 The Justice Mechanism

Definition 5.2 (Justice Mechanism)**.** *A* justice mechanism J *is a set of institutional operations—courts, elections, redistribution, reform, debt cancellation, revolution—that reduce the injustice gap. Formally,* J *produces a rate* $j(G) > 0$ *at which the gap is reduced:*

$$\left.\frac{dG}{dt}\right|_{justice} = -j(G). \tag{10}$$

In plain language: justice is the machinery that pushes back against drift. Courts, elections, redistribution, debt cancellation—these are all ways of rebalancing the ledger before it breaks. They are not acts of charity; they are the institutional operations that keep the ledger from drifting too far out of balance.

5.3 The Dynamical System

Combining drift and justice, the injustice gap evolves according to

$$\frac{dG}{dt} = d(G) - j(G). \tag{11}$$

Proposition 5.1 (Effectiveness of Justice)**.** *A justice mechanism J is* effective *if and only if it reduces the injustice gap faster than drift increases it:*

$$j(G) > d(G) \quad \textit{for all } G \in [0, \bar{G}]. \tag{12}$$

An effective justice mechanism keeps the order within the viability set; an ineffective one does not.

Proof. From (11), $dG/dt = d(G) - j(G)$. If $j(G) > d(G)$ for all G, then $dG/dt < 0$ throughout, and the gap decreases toward zero. If $j(G) < d(G)$ for all G, then $dG/dt > 0$ throughout, and the gap grows without bound, eventually exceeding $\bar{G}$ and rupturing the order. If the two rates cross, the crossing point is an unstable equilibrium: below it the gap decreases, above it the gap grows. □

In plain language: a justice mechanism works if it rebalances the ledger faster than the ledger drifts out of balance. If it can keep up, the order stays stable. If it cannot, the order eventually collapses.

Corollary 5.1 (Collapse Mode)**.** *Regimes with effective justice mechanisms collapse gradually or renew; regimes without them collapse suddenly. The dynastic cycle, the electoral defeat, and the regime change are all forms of the same operation—the rebalancing of the ledger—differing only in whether the rebalancing is incremental or catastrophic.*

Proposition 5.2 (Oscillation)**.** *If the justice mechanism is effective only up to a threshold—$j(G) > d(G)$ for $G < G^*$ and $j(G) < d(G)$ for $G > G^*$—then the system oscillates around G^* rather than drifting away from it. This is the dynastic cycle: concentration, decay, rebellion, renewal, and concentration again.*

Proof. At $G < G^*$, justice outruns drift, so the gap decreases. At $G > G^*$, drift outruns justice, so the gap increases. The system is therefore pulled back toward G^* from both sides, producing oscillation around the threshold. The oscillation is the dynastic cycle: the order drifts toward concentration until the gap crosses G^*, at which point the justice mechanism (rebellion, reform, revolution) rebalances it, and the cycle repeats. □

In plain language: this is the dynastic cycle—concentration of wealth, decay of legitimacy, rebellion, renewal, and then concentration again. It has happened hundreds of times across thousands of years, on every continent. The details vary. The pattern does not.

6 The Extraction Ceiling

The extraction ceiling is the maximum amount a principal can extract from an agent, and it is bounded by the agent's exit cost. Consider a relation e between a principal p and an agent a.

Assumption 6.1 (The Participation Constraint)**.** *The agent a remains in relation e if and only if their net gain from staying is at least their effective outside option:*

$$n_a^e \geq u_a - k_a^e, \tag{13}$$

where u_a is the agent's outside option (the value of their best alternative) and k_a^e is the cost of exercising that option (the exit cost).

The participation constraint says that the agent stays if staying is at least as good as leaving. The exit cost k_a^e enters negatively: the higher the cost of leaving, the lower the effective outside option, and the more the agent will tolerate.

Proposition 6.1 (The Extraction Ceiling)**.** *The maximum contribution the principal can demand from the agent is*

$$c_a^{e,\max} = s_a^e B^e - u_a + k_a^e, \tag{14}$$

which is strictly increasing in the agent's exit cost k_a^e. The principal's power over the agent is bounded by the agent's exit cost.

Proof. The principal maximizes their own net gain $n_p^e = s_p^e B^e - c_p^e$ subject to the agent's participation constraint (6.1). The principal can raise the agent's contribution c_a^e (or lower the agent's share s_a^e) until the constraint binds: $n_a^e = u_a - k_a^e$. Substituting $n_a^e = s_a^e B^e - c_a^e$ and solving for c_a^e gives $c_a^{e,\max} = s_a^e B^e - u_a + k_a^e$. Differentiating with respect to k_a^e gives $\partial c_a^{e,\max}/\partial k_a^e = 1 > 0$. The ceiling is increasing in the exit cost. □

In plain language: the more trapped someone is, the more can be taken from them. A worker who cannot leave a bad job can be squeezed harder than one who has five offers. The ceiling on extraction is set by how hard it is to walk away.

Corollary 6.1 (Exit Asymmetry Determines the Ceiling)**.** *Regimes that concentrate exit costs on producers and voice on elites have higher extraction ceilings than regimes that distribute them evenly. The asymmetry of exit costs is the structural precondition of every other extraction.*

Proof. By Proposition 6.1, the extraction ceiling on each actor is increasing in that actor's exit cost. If producers have high exit costs and elites have low exit costs, the ceiling on producers is high and the ceiling on elites is low, so the regime can extract more from producers. If exit costs are equalized, the ceilings are equalized, and the total extraction falls. □

Corollary 6.2 (The Deepest Lever)**.** *Equalizing exit costs and expanding voice are the deepest levers of reform, because they change the ceiling rather than the level of extraction. Redistribution without equalizing exit costs treats the symptom; the illness is structural.*

7 The Patience Threshold

The patience threshold concerns the conservation of hosts. A host is a resource stock that a principal controls and can either conserve or consume. The principal's choice depends on their discount rate.

Assumption 7.1 (The Renewable Resource)**.** *A host is a stock $K \in [0, 1]$ (normalized) that grows logistically: $g(K) = K(1 - K)$. The principal harvests h per period, so the stock evolves as*

$$K_{t+1} = K_t + K_t(1 - K_t) - h_t. \tag{15}$$

The principal discounts the future at rate r, with discount factor $\delta = 1/(1+r)$, and maximizes the present value of harvests:

$$\max_{\{h_t\}} \sum_{t=0}^{\infty} \delta^t h_t. \tag{16}$$

Proposition 7.1 (The Patience Threshold)**.** *The optimal steady-state host stock is*

$$K^*(r) = \frac{1-r}{2}, \tag{17}$$

which is strictly decreasing in the discount rate r. A patient principal (low r) conserves the host; an impatient principal (high r) consumes it.

Proof. In steady state, the harvest equals the growth: $h = g(K) = K(1-K)$. The principal's Bellman equation is

$$V(K) = \max_h \{h + \delta\, V(K + g(K) - h)\}. \tag{18}$$

The first-order condition is $1 = \delta V'(K')$, where $K' = K + g(K) - h$. The envelope condition is $V'(K) = \delta V'(K')\,(1 + g'(K))$. In steady state $K' = K$, so combining the two conditions gives $1 = \delta(1 + g'(K))$. Substituting $\delta = 1/(1+r)$ and solving gives $g'(K^*) = r$. With $g(K) = K(1-K)$, we have $g'(K) = 1 - 2K$, so $1 - 2K^* = r$, hence $K^*(r) = (1-r)/2$. Differentiating, $dK^*/dr = -1/2 < 0$. □

In plain language: the longer you expect to be around, the more you take care of what you have. A dynasty that expects to last a thousand years invests in infrastructure. A junta facing the next coup loots the treasury. Patience is the difference between conserving and consuming.

Corollary 7.1 (Horizon Determines Patience)**.** *A government with a long time horizon has a low discount rate and conserves its hosts; a government with a short time horizon has a high discount rate and consumes them. The patience of a government is determined by its time horizon, not by its ideology.*

Proof. The discount rate r is the inverse of the time horizon: a government that expects to last a long time discounts the future lightly (low r), while a government that expects to be replaced soon discounts it heavily (high r). By Proposition 7.1, the host stock is decreasing in r, so a long-horizon government conserves and a short-horizon government consumes. □

Corollary 7.2 (The Conservation Principle Fails Under Uniqueness)**.** *When the host is unique and non-substitutable, the conservation principle fails. The principal cannot afford to be patient, because patience means losing to rivals who are impatient. The result is a race to consume the host before someone else does.*

Proof. The derivation of $K^*(r)$ assumes the principal faces no competition for the host. With a unique host and competing principals, each principal's effective discount rate rises, because the value of conserving the host is captured by whichever rival consumes it first. The race to consume is the rational response to a unique, non-substitutable host. □

8 The Legitimacy Constraint

Definition 8.1 (Legitimacy)**.** Legitimacy $L \in [0,1]$ *is the degree to which the population believes that the existing distribution of the ledger and the authority of the government are rightful, appropriate, or justified.*

Definition 8.2 (Cost of Governance). *The* cost of governance $\Phi(G, L)$ *is the resources—administrative, coercive, economic—required to achieve a given level of compliance with the ledger at injustice gap* G *and legitimacy* L*.*

Assumption 8.1 (Compliance Requires Coercion in Proportion to Illegitimacy). *To achieve a required level of compliance* $c(G)$*, the government must coerce the fraction of the population that does not comply voluntarily. The voluntarily compliant fraction is* L*, so the coerced fraction is* $1 - L$*. The cost of coercion is convex in the amount coerced.*

Proposition 8.1 (The Legitimacy Constraint). *The cost of governance* $\Phi(G, L)$ *is strictly decreasing in legitimacy* L*:*

$$\frac{\partial \Phi}{\partial L} < 0. \tag{19}$$

As legitimacy falls, the government must substitute repression for compliance, and repression is more expensive.

Proof. By Assumption 8.1, the cost of governance is $\Phi(G, L) = \phi\big((1 - L)\, c(G)\big)$, where ϕ is a convex cost function and $c(G)$ is the compliance required at gap G. Differentiating with respect to L:

$$\frac{\partial \Phi}{\partial L} = \phi'\big((1 - L)\, c(G)\big) \cdot \big(-c(G)\big) < 0, \tag{20}$$

since $\phi' > 0$ and $c(G) > 0$. The cost is strictly decreasing in legitimacy. □

In plain language: it is cheaper to be fair than to police unfairness. When people believe the order is just, they comply voluntarily. When they do not, you have to force them—and force is expensive.

Corollary 8.1 (Legitimacy Is a Fiscal Variable). *Legitimacy is not a moral ornament. It is a fiscal and administrative variable. A government's legitimacy is measurable by the cost of achieving a given level of compliance.*

Corollary 8.2 (Maximum Sustainable Injustice). *A legitimacy regime that is accepted by the population allows the coalition to extract more, because the cost of compliance is lower. The maximum sustainable injustice is increasing in legitimacy.*

Proof. The coalition can extract up to the point where the cost of governance becomes prohibitive. By Proposition 8.1, higher legitimacy lowers the cost of governance at any gap, so the coalition can sustain a larger gap before the cost becomes prohibitive. The maximum sustainable injustice is therefore increasing in legitimacy. □

9 The Planner's Problem and the Weighting Vector

Definition 9.1 (The Planner's Problem). *A government implements a governance map* G *that maximizes a weighted sum of actors' net gains:*

$$\max_{G} \sum_{i \in N} \lambda_i\, n_i(G), \tag{21}$$

subject to the ledger constraints, where $\lambda_i \geq 0$ *are weights with* $\sum_i \lambda_i = 1$*. The weighting vector* $\lambda = (\lambda_1, \ldots, \lambda_n)$ *encodes whose interests the government serves.*

Proposition 9.1 (Government Is a Coalitional Instrument)**.** *The distribution of the ledger under a government is determined by the weighting vector λ, which is set by the coalition that controls the government. Different coalitions implement different λ, hence different distributions, regardless of constitutional form.*

Proof. The planner's problem (9.1) has a solution $G^*(\lambda)$ that depends on the weights. An actor i with $\lambda_i = 0$ receives no weight in the objective, so the solution does not serve their interests. An actor with large λ_i is served. The weights are set by the coalition that controls the apparatus: the coalition assigns positive weight to its own members and zero (or small) weight to others. The distribution therefore follows the coalition, not the constitutional form. □

In plain language: every government serves someone. The question is who. The weights in the government's calculation are set by the coalition that keeps it in power—and the coalition is usually a lot smaller than the population.

Corollary 9.1 (Testability of the Claim to Govern for All)**.** *A government's claim to govern for all is testable. It is true if the realized G moves the ledger toward the viability set for all actors—equivalently, if λ_i is positive for all actors whose capacity is at stake. It is false if G moves the ledger toward viability for the coalition at the expense of others.*

Corollary 9.2 (Justice as Gap Reduction for the Coalition)**.** *The justice mechanism reduces the injustice gap for the coalition, not for the population. The partiality of the coalition is the partiality of justice.*

Proof. By Proposition 9.1, the government serves the coalition's weights. The justice mechanism, as an operation of the government, reduces the gap for the actors with positive weight—the coalition—and leaves the gap for others unaddressed. Justice is partial to the extent that the coalition is partial. □

10 The Core Propositions Collected

The companion papers cite the following results, all derived above.

1. **The classification of relations** (Definition 3.1): symbiotic, parasitic, and predatory, distinguished by the sign and sustainability of net transfers.
2. **The extraction ceiling** (Proposition 6.1): the principal's power is bounded by the agent's exit cost; the ceiling is increasing in exit cost.
3. **The patience threshold** (Proposition 7.1): the optimal host stock is $K^*(r) = (1 - r)/2$, decreasing in the discount rate.
4. **The legitimacy constraint** (Proposition 8.1): the cost of governance is decreasing in legitimacy.
5. **The coalitional instrument** (Proposition 9.1): the distribution is set by the weighting vector λ_i, which the coalition controls.
6. **The dynamical system** (Proposition 5.1): the order is stable when justice outruns drift, oscillates when they balance, and collapses when drift outruns justice.

Two further results, used by the companion papers on voice and bargaining, follow directly from the framework.

Proposition 10.1 (Atomization Prevents Aggregation)**.** *The aggregation of grievance is increasing in the coordination capacity of the actors, which is a function of voice and connectivity. Atomization—the isolation of actors from one another—reduces coordination capacity, preventing the aggregation of grievance.*

Proof. Grievance aggregates when actors can coordinate their claims. Coordination requires communication and organization, which are functions of voice v_i^e and of the connectivity of the network. Atomization reduces both: it suppresses voice and severs the connections through which claims could be coordinated. The result is that grievance remains individualized and cannot aggregate into collective claims. □

In plain language: a system can be deeply unjust while appearing peaceful, if people believe "it is just me" rather than "it is all of us." Atomization is often more effective than censorship at suppressing politics.

Proposition 10.2 (Voice and Exit Skew the Bargaining Outcome)**.** *The bargaining outcome between two parties is skewed toward the party with higher voice and lower exit cost. Asymmetries in voice and exit skew the outcome.*

Proof. The Nash bargaining solution maximizes $(n_i - d_i)(n_j - d_j)$, where $d_i = u_i - k_i$ is the disagreement point (the effective outside option). A party with a high exit cost k_i has a low disagreement point d_i, so the solution is skewed against them. A party with high voice v_i can influence the terms directly, skewing the solution toward them. Both asymmetries move the outcome in the same direction: toward the party with low exit cost and high voice. □

In plain language: in any negotiation, the person who can walk away and the person who can speak loudly get the better deal. The person who is trapped and silenced gets the worse one.

11 Conclusion

Civilization is a ledger, not a monument. Art, architecture, law, science, and engineering are outcomes of the relations that produce them, not the relations themselves. Every stable order solves a reciprocity problem: people must be able to reproduce their capacity to participate. When they cannot, they resist. When resistance is suppressed, the pressure builds. When the pressure breaks through, the order collapses.

The framework derived here is a discipline of attention, not a total explanation. It tells the reader what to look at: the ledger, the classification of relations, the exit costs, the voice, the injustice gap, the drift, and the justice mechanism. It derives the five propositions on which the companion papers depend, and it makes each derivation inspectable.

The framework does not tell the reader what to value. It tells the reader what to look at, and it asks the question that every order must answer, and that most orders conceal:

Whose ledger is this, and for whom?

The question is not answered by the framework. It is made possible by the framework. The answer is the work of those who ask it.

12 References

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